TinkerSub KB-1 side plates: templates for one 24 x 24 in HDPE sheet, using two factory edges per plate

Each plate's skid (bottom) edge and front end sit on factory edges of the sheet: those are the GREEN lines. Don't cut them.

Check first: the sheet's two diagonals must match within about 2 mm (square corners). If not, use the standard templates instead.

1 Print every page at 100 % / Actual size, NOT 'Scale to fit'. Measure the 100 mm bar on each page: exactly 100 mm.

2 Each plate is 4 tiles, A1 A2 / B1 B2 (map in each tile's corner). Cut every tile on its black box line, all four sides. Butt the edges together, A1 beside A2 and above B1, so the tick marks and the drawing run straight across the joint, and tape on the back.

3 Trim the taped template EXACTLY on its two green lines, so the paper edge is the plate edge. The rest of the outline can stay untrimmed.

4 Lay it on the sheet with its green edges flush against the sheet edges (layout on the right): PORT in the bottom-left corner, STARBOARD in the top-left corner. Starboard is printed turned end for end (skid at the top): it is still outside face up. Glue down (3M Super 77 or glue stick). Slide a straightedge along each green edge to check it is flush the whole way.

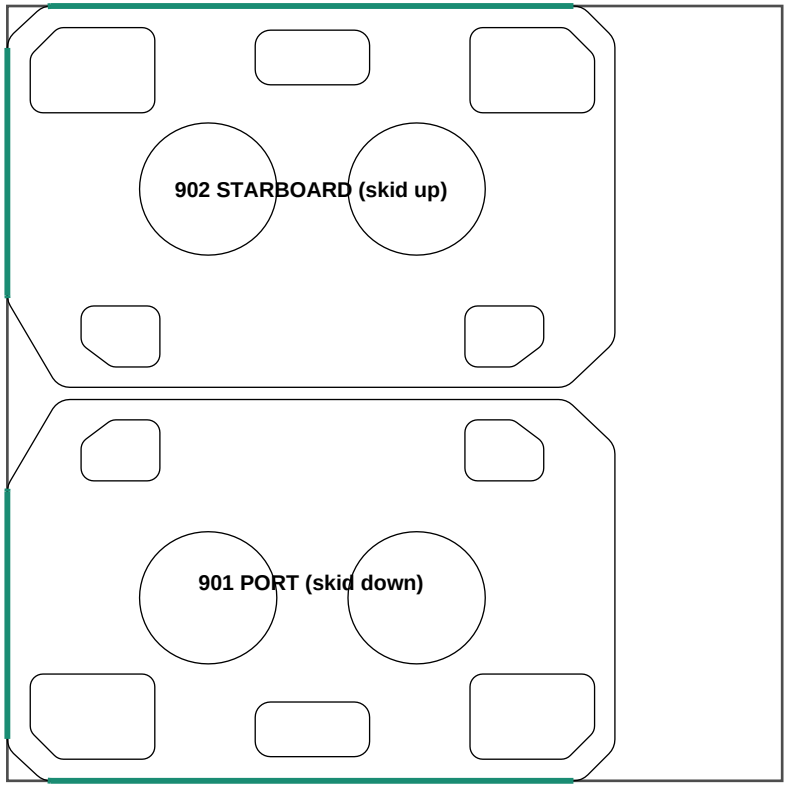
5 Centre-punch every crosshair. Drill 3.5 mm (M3), 4.5 mm (M4), 5.5 mm (M5) as printed, slow, on a scrap-wood backer.

6 Drill each blue 10 mm starter hole, jigsaw the windows, then the RED outline only: the two top edges (9.6 mm waste strip between them: one cut just outside each line), the rear ends and the corners. Fine blade, slow speed, just outside the line.

7 Sand the cut edges, break the corners, mark 901 PORT / 902 STBD and the hole tags on the outside faces, peel the paper off.

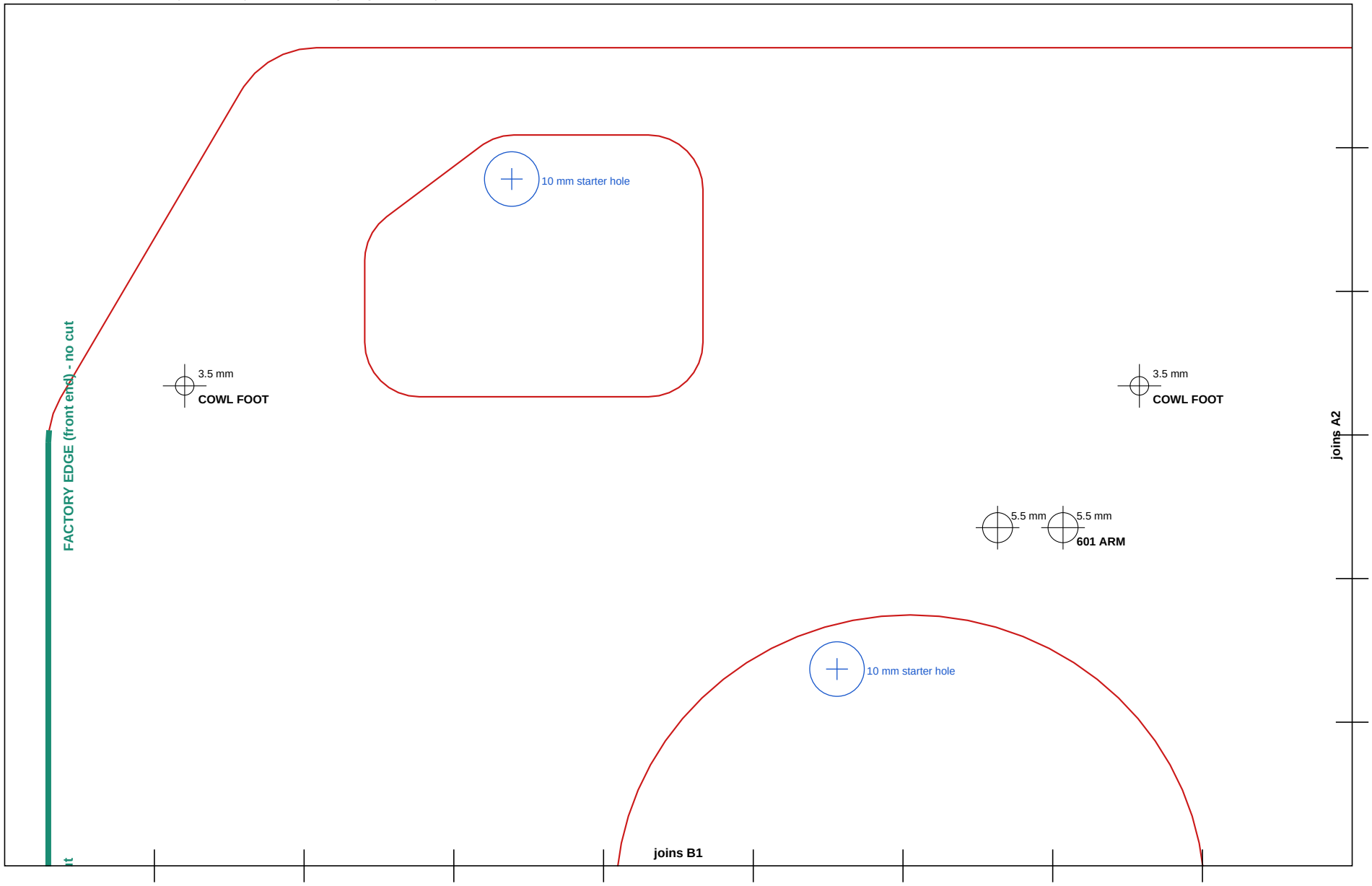
A 131 x 610 mm offcut is left at the right-hand side of the sheet: keep it (crossbar, brackets).

green = factory edge, no cut (front ends of both plates on the left edge)



24 x 24 in sheet (609.6 mm), outside faces up

CUT ON THE BLACK BOX LINE (all four sides), then butt the edges together and tape on the back



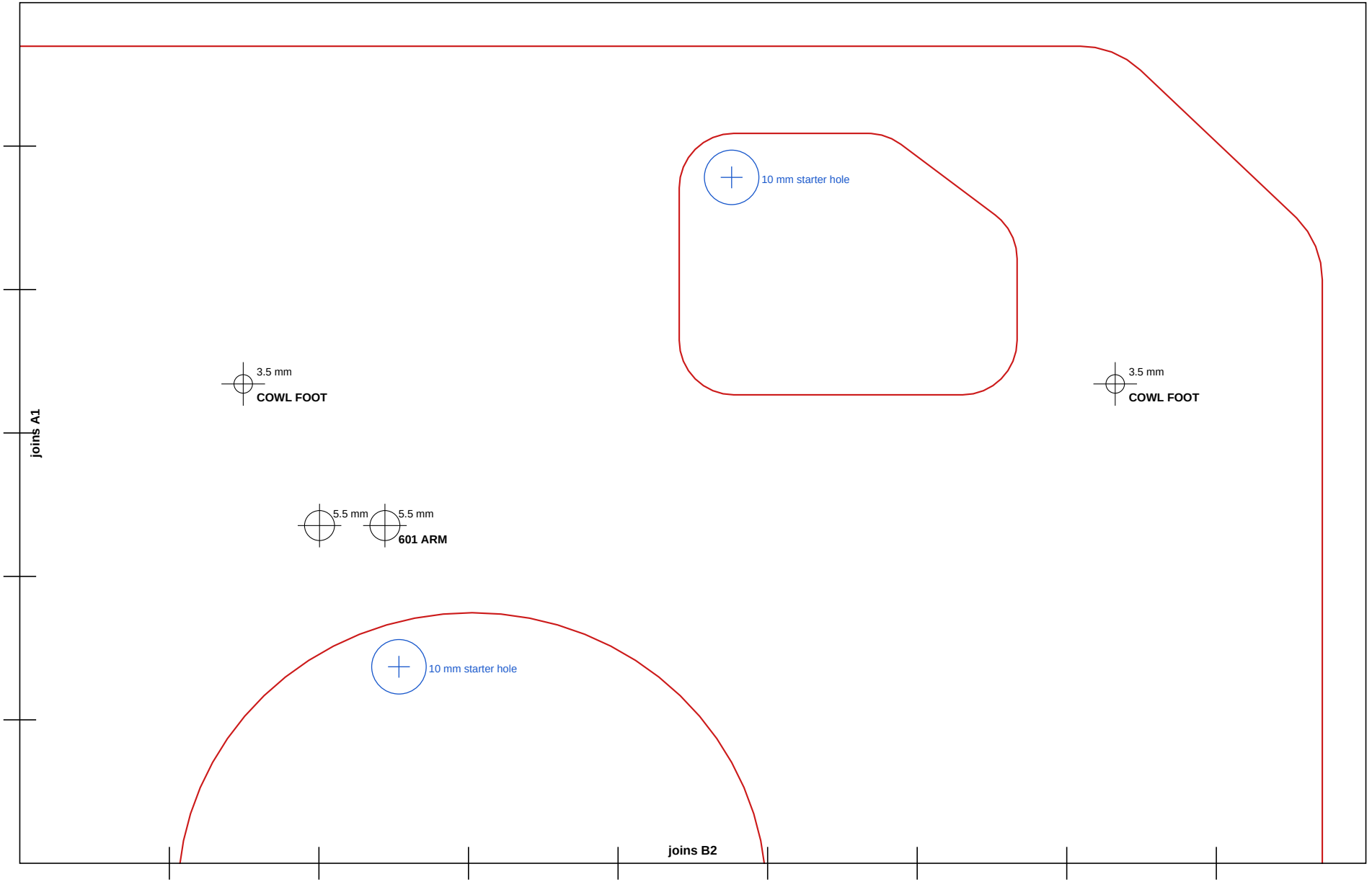
100 mm

901 PORT (skid DOWN) - tile A1 of 4 - outside face up, front of the ROV on the LEFT

100 %, no scaling; check the 100 mm bar. Green = factory edge (no cut, trim paper flush later). Red = cut in the HDPE. 3.5 mm = M3, 4.5 = M4, 5.5 = M5.
Paper: cut each tile on its black box, butt the edges so the ticks and lines meet, tape on the back.

A1	A2
B1	B2

CUT ON THE BLACK BOX LINE (all four sides), then butt the edges together and tape on the back

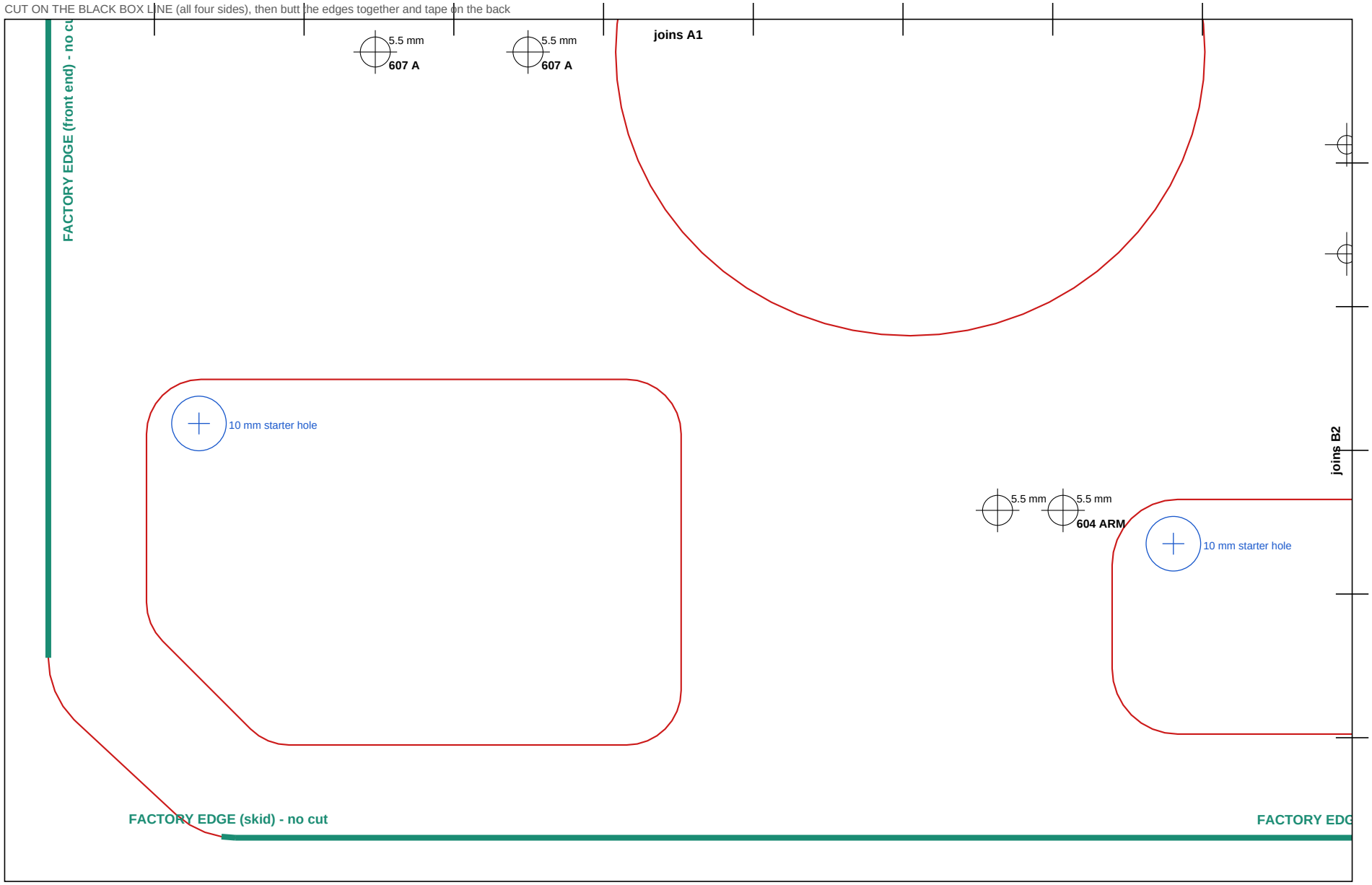


100 mm

901 PORT (skid DOWN) - tile A2 of 4 - outside face up, front of the ROV on the LEFT

100 %, no scaling; check the 100 mm bar. Green = factory edge (no cut, trim paper flush later). Red = cut in the HDPE. 3.5 mm = M3, 4.5 = M4, 5.5 = M5.
Paper: cut each tile on its black box, butt the edges so the ticks and lines meet, tape on the back.

A1	**A2**
B1	B2

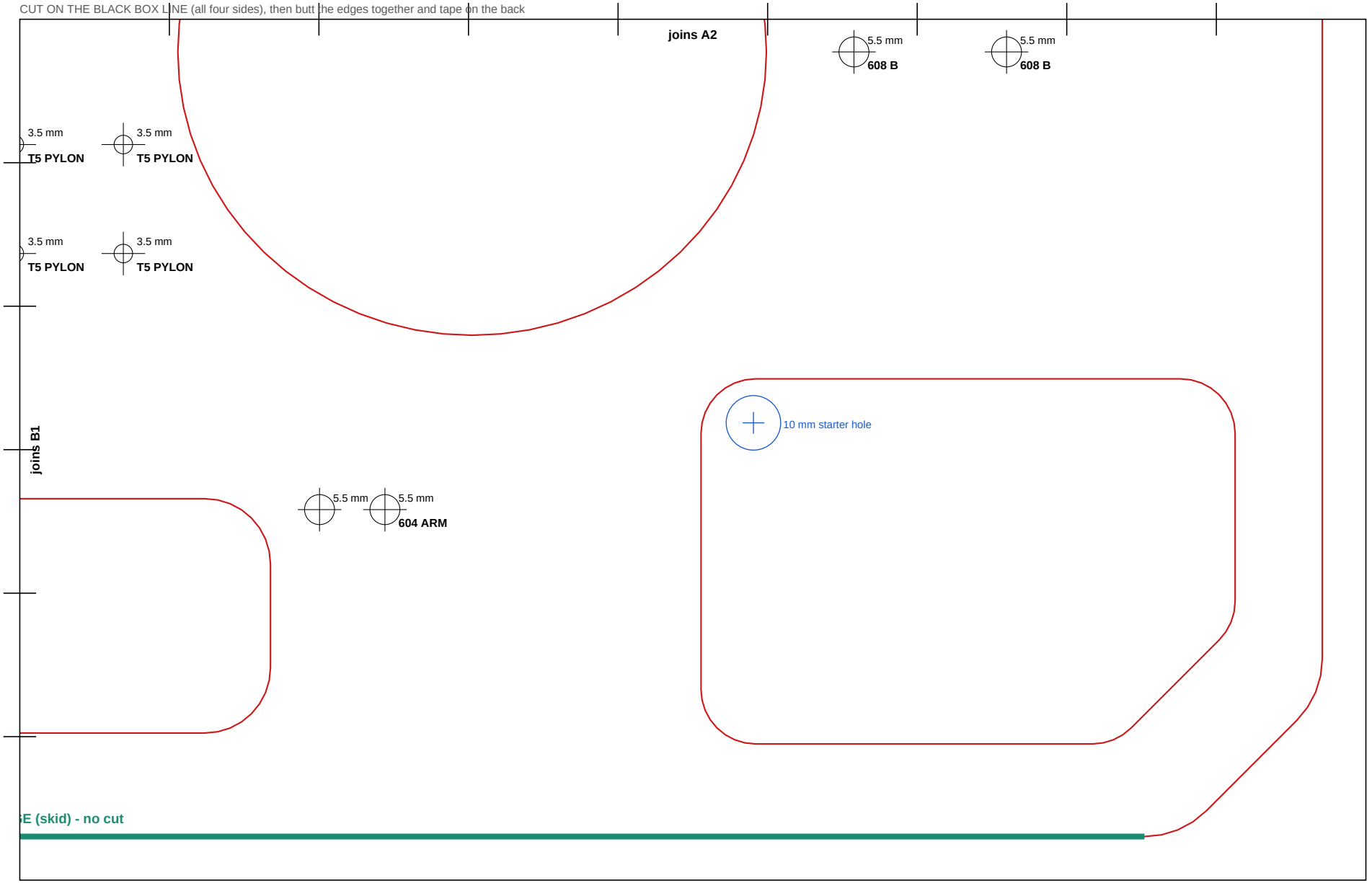


100 mm

901 PORT (skid DOWN) - tile B1 of 4 - outside face up, front of the ROV on the LEFT

100 %, no scaling; check the 100 mm bar. Green = factory edge (no cut, trim paper flush later). Red = cut in the HDPE. 3.5 mm = M3, 4.5 = M4, 5.5 = M5.
Paper: cut each tile on its black box, butt the edges so the ticks and lines meet, tape on the back.

A1	A2
B1	B2



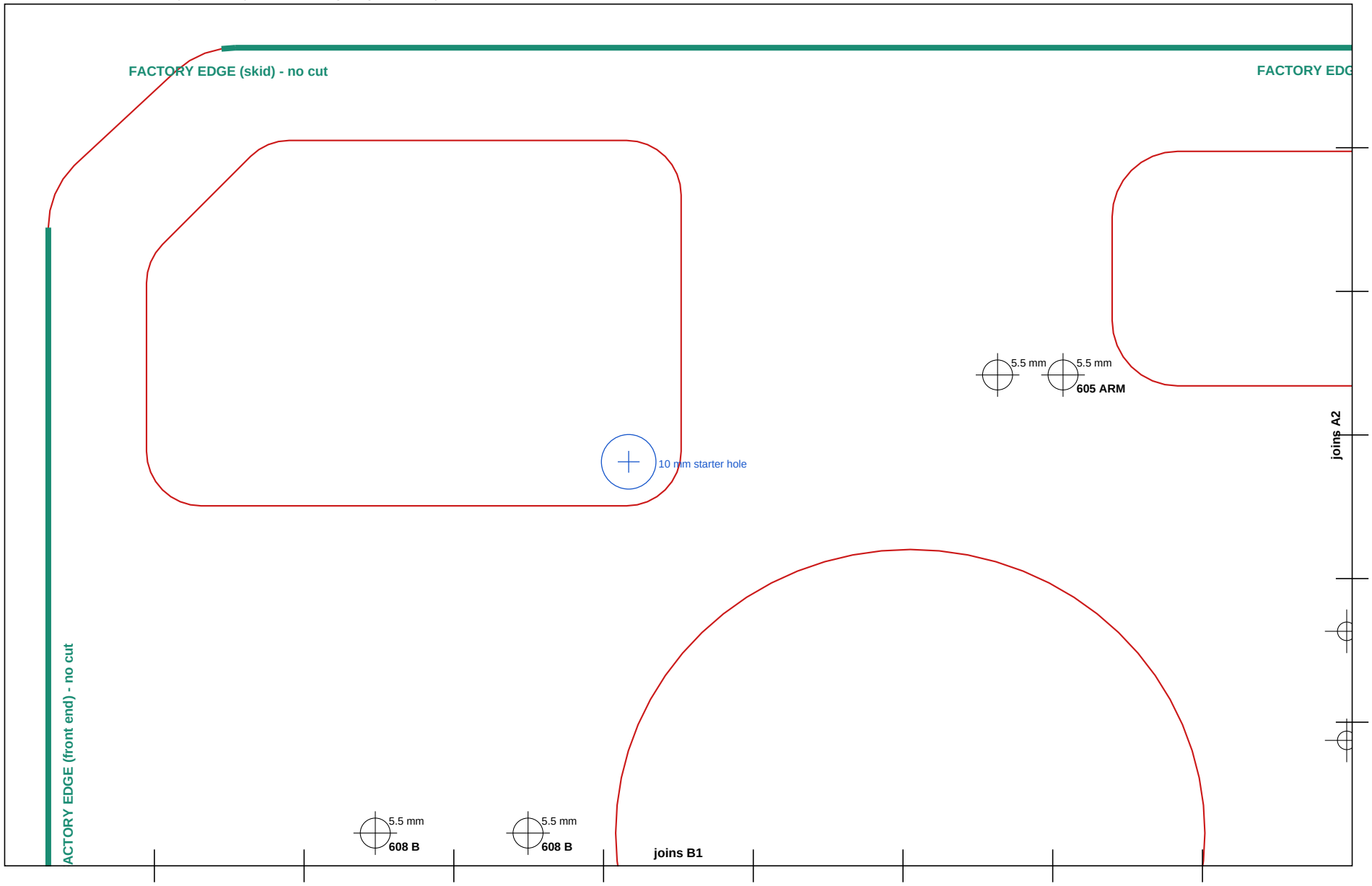
100 mm

901 PORT (skid DOWN) - tile B2 of 4 - outside face up, front of the ROV on the LEFT

100 %, no scaling; check the 100 mm bar. Green = factory edge (no cut, trim paper flush later). Red = cut in the HDPE. 3.5 mm = M3, 4.5 = M4, 5.5 = M5.
Paper: cut each tile on its black box, butt the edges so the ticks and lines meet, tape on the back.

A1	A2
B1	B2

CUT ON THE BLACK BOX LINE (all four sides), then butt the edges together and tape on the back

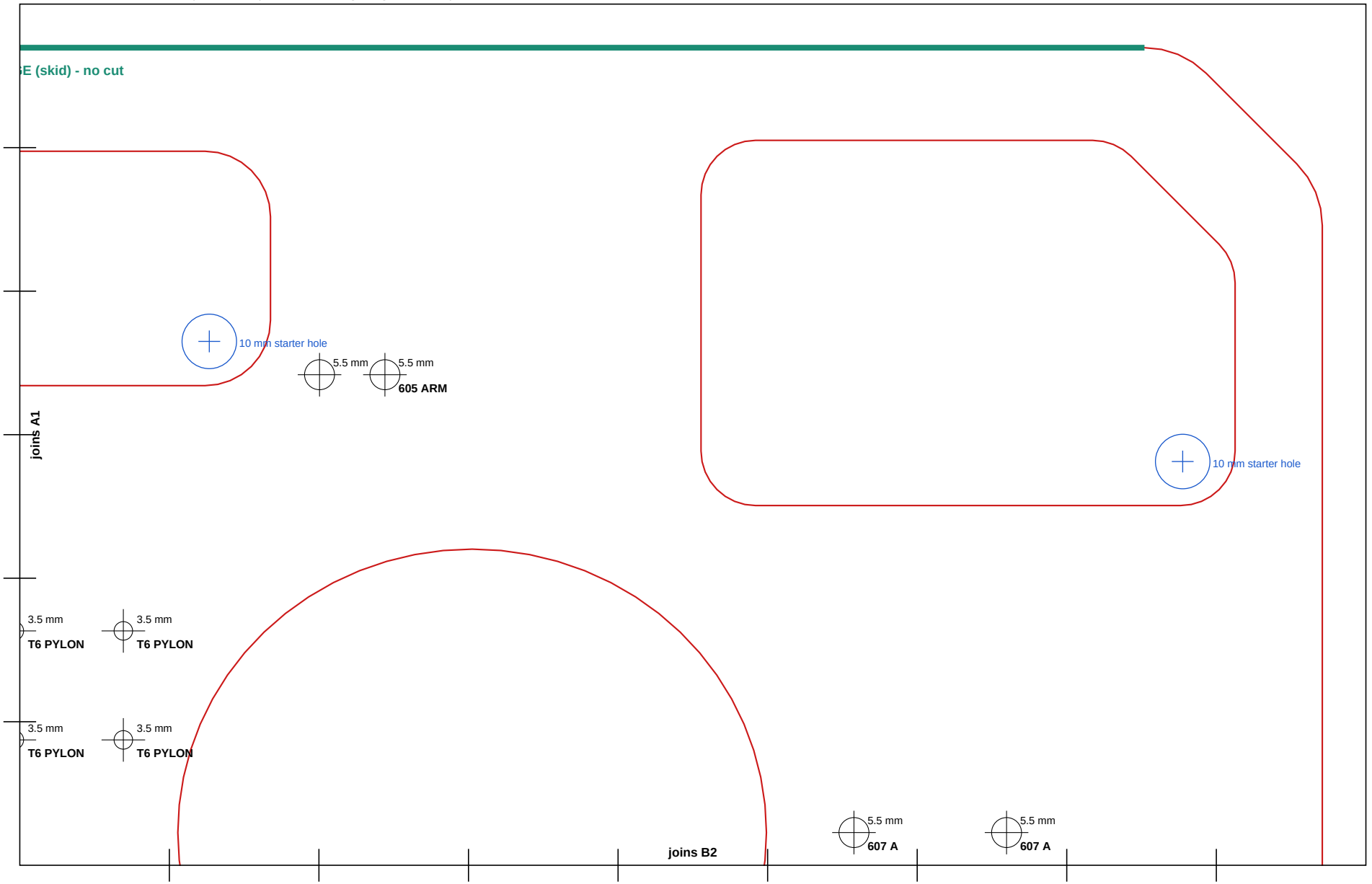


902 STARBOARD (turned end for end: skid UP) - tile A1 of 4 - outside face up, front of the ROV on the LEFT

100 %, no scaling; check the 100 mm bar. Green = factory edge (no cut, trim paper flush later). Red = cut in the HDPE. 3.5 mm = M3, 4.5 = M4, 5.5 = M5.
Paper: cut each tile on its black box, butt the edges so the ticks and lines meet, tape on the back.

A1	A2
B1	B2

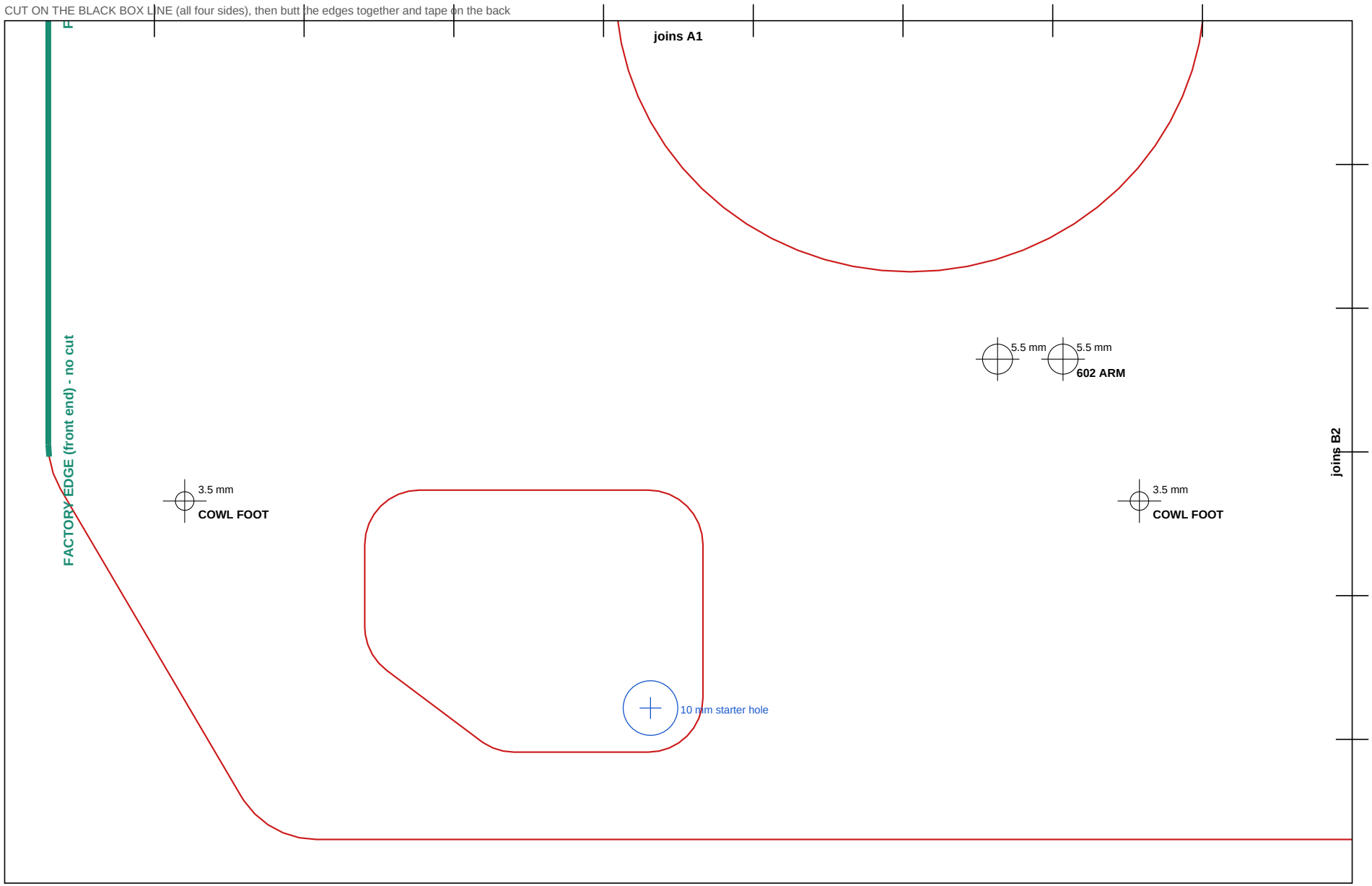
CUT ON THE BLACK BOX LINE (all four sides), then butt the edges together and tape on the back



902 STARBOARD (turned end for end: skid UP) - tile A2 of 4 - outside face up, front of the ROV on the LEFT

100 %, no scaling; check the 100 mm bar. Green = factory edge (no cut, trim paper flush later). Red = cut in the HDPE. 3.5 mm = M3, 4.5 = M4, 5.5 = M5.
Paper: cut each tile on its black box, butt the edges so the ticks and lines meet, tape on the back.

A1	**A2**
B1	B2

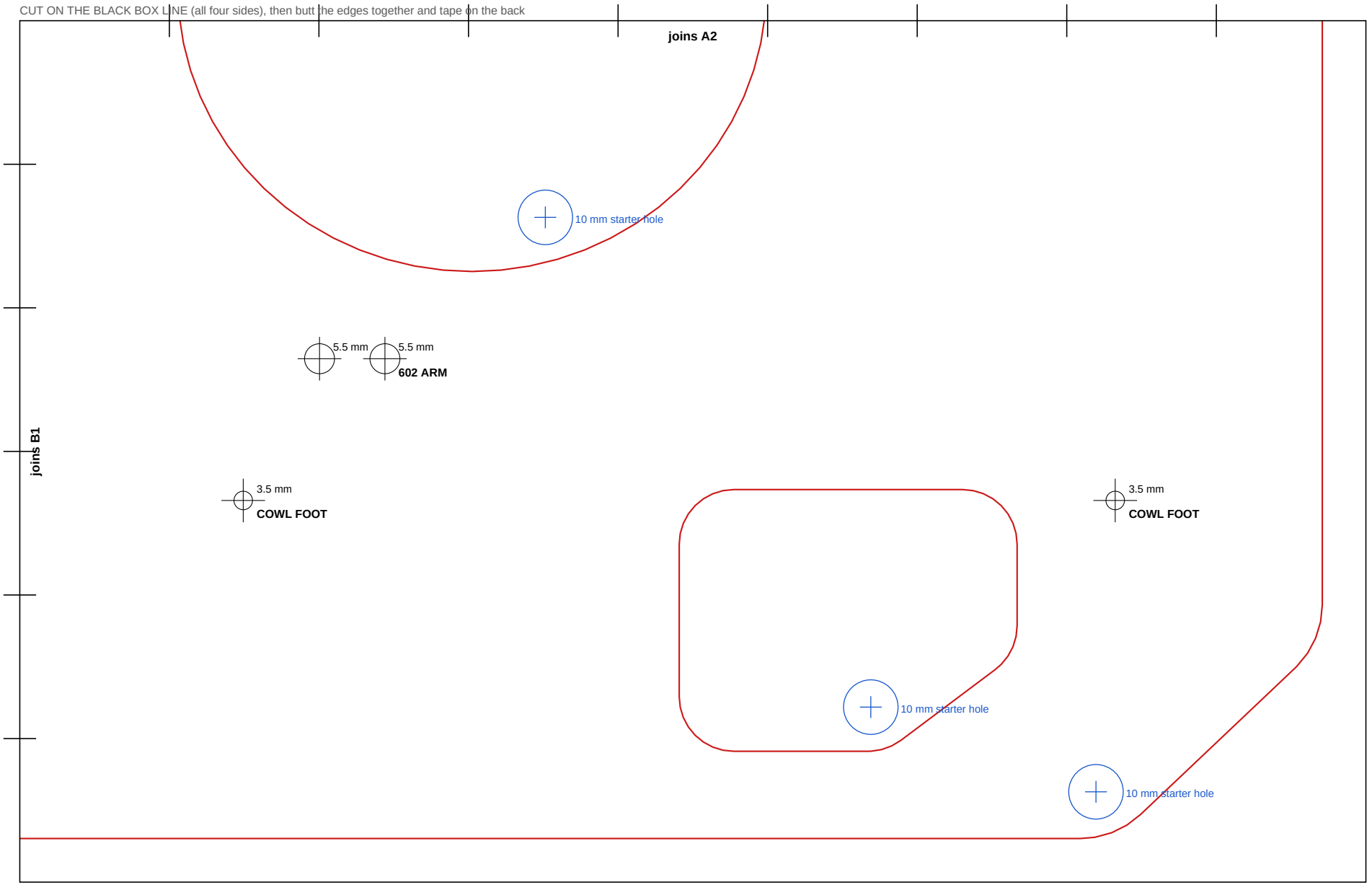


902 STARBOARD (turned end for end: skid UP) - tile B1 of 4 - outside face up, front of the ROV on the LEFT

100 %, no scaling; check the 100 mm bar. Green = factory edge (no cut, trim paper flush later). Red = cut in the HDPE. 3.5 mm = M3, 4.5 = M4, 5.5 = M5.

Paper: cut each tile on its black box, butt the edges so the ticks and lines meet, tape on the back.

A1	A2
B1	B2



902 STARBOARD (turned end for end: skid UP) - tile B2 of 4 - outside face up, front of the ROV on the LEFT

100 %, no scaling; check the 100 mm bar. Green = factory edge (no cut, trim paper flush later). Red = cut in the HDPE. 3.5 mm = M3, 4.5 = M4, 5.5 = M5.
Paper: cut each tile on its black box, butt the edges so the ticks and lines meet, tape on the back.

A1	A2
B1	**B2**